

Draft Introduction and Methods

I. Introduction

Magnetic shielding is essential for instruments that require a low-field and low-noise environment, including optically pumped magnetometers, spin-exchange-relaxation-free (SERF) magnetometers, comagnetometers, magnetoencephalography systems, wireless power transfer devices, and precision measurements in fundamental physics [1]-[6]. In these applications, external static and low-frequency magnetic disturbances are commonly suppressed by passive magnetic shielding, where high-permeability materials provide magnetic-flux shunting and conductive materials contribute eddy-current attenuation [7]-[9]. Multilayer magnetic shielding rooms and compact shielded chambers made of permalloy have achieved residual magnetic fields at the nanotesla level or below after demagnetization [10]-[12]. However, as magnetic sensors continue to approach femtotesla and sub-femtotesla sensitivity, the magnetic shield itself becomes an important noise source rather than a passive background component [13], [14].

Conventional low-frequency magnetic shields are usually fabricated from permalloy, aluminum, copper, steel, or their laminated combinations [15]-[18]. These materials are effective for field attenuation, but their intrinsic loss mechanisms generate magnetic noise. In metallic shields, the relatively high electrical conductivity produces eddy-current magnetic noise, which is particularly unfavorable for SERF magnetometers and other ultra-sensitive sensors operated below several hundred hertz [13], [14]. Ferrite materials, especially Mn-Zn ferrites, provide an attractive alternative because they combine high magnetic permeability with electrical conductivity several orders of magnitude lower than that of permalloy [19]. Previous studies have shown that ferrite shields can significantly reduce eddy-current noise and that the remaining low-frequency noise is mainly associated with hysteresis loss [20]-[23].

Although ferrite magnetic shields have been widely studied in cylindrical, annular, and cubic configurations, practical implementation is still constrained by material processing and structural assembly [20], [23], [24]. Thick monolithic ferrite shells are difficult to manufacture, fragile during machining, and limited in size. Ferrite sheets or strips are easier to fabricate, but the resulting shield is no longer an ideal continuous shell. The layer thickness, radial gap, circumferential segmentation, and total ferrite volume all influence both the shielding factor and the magnetic-noise contribution. Therefore, a design strategy based only on increasing material thickness is insufficient. A quantitative model is needed to evaluate whether multilayer thin ferrite structures can improve shielding efficiency and reduce magnetic noise under realistic manufacturability constraints.

The magnetic noise of a shield can be calculated from the dissipated power in the magnetic material by using the fluctuation-dissipation theorem [25]-[28]. This method makes it possible to evaluate complex structures through finite-element analysis (FEA), because the magnetic-noise indicator can be expressed as an integral of the squared magnetic field strength over the material volume [28], [29]. For ferrites at low frequency, hysteresis loss dominates over eddy-current loss, and the noise is closely related to the imaginary part of the complex permeability and the spatial distribution of the magnetic field inside each layer [13], [22], [29]. Layer-resolved integration is therefore important for multilayer shields, since different layers may contribute unequally to the total noise.

In this article, a multilayer cylindrical thin-ferrite magnetic shield is modeled and evaluated for low-noise magnetic shielding applications. The proposed structure consists of multiple ferrite layers separated by radial air or adhesive gaps. The study focuses on manufacturable ferrite sheets with a single-layer thickness of 0.08-0.60 mm and compares single-layer, multilayer continuous-shell, and segmented-shell configurations. The shielding factor, residual center field, total ferrite volume, and magnetic-noise-related integral are used as the main evaluation metrics. The objective is to determine whether multilayer thin ferrite shells can achieve a favorable tradeoff between magnetic shielding performance, noise suppression, and material usage.

The remainder of this article is organized as follows. Section II introduces the magnetic-noise calculation model, the multilayer cylindrical-shell geometry, and the finite-element evaluation method. Section III presents the simulation results for continuous-shell screening and segmented-shell correction. Section IV discusses the relationship among layer number, ferrite thickness, radial gap, shielding factor, and magnetic-noise indicator. Section V summarizes the conclusions and the implications for the design of manufacturable low-noise ferrite magnetic shields.

II. Modeling and Analysis

A. Magnetic-Noise Model of a Ferrite Magnetic Shield

The magnetic noise generated by a passive shield originates from energy dissipation in the shielding material. Following the fluctuation-dissipation theorem, the magnetic field noise at an observation point can be related to the power dissipated in the shield when a virtual pickup coil is placed at that point [25], [28]. If a coil with area A , turn number N , and angular frequency ω detects a magnetic-field fluctuation with power spectral density $S_B(f)$, the corresponding voltage noise is

$$S_V(f) = A^2 N^2 \omega^2 S_B(f).$$

The voltage fluctuation is also related to the effective resistance $R_{\text{eff}}(f)$ by

$$S_V(f) = 4kTR_{\text{eff}}(f),$$

where k is the Boltzmann constant and T is the absolute temperature. For a sinusoidal current $I(t) = I \sin \omega t$, the time-averaged dissipated power is

$$P(f) = \frac{1}{2} I^2 R_{\text{eff}}(f).$$

Combining the above equations gives the magnetic field noise amplitude spectral density

$$\delta B(f) = \sqrt{S_B(f)} = \frac{1}{ANI\omega} \sqrt{8kTP(f)}.$$

For a high-permeability ferrite shield, the dissipated power contains eddy-current loss and hysteresis loss. They can be written as

$$P_{\text{eddy}} = \int_V \frac{1}{2} \sigma E^2 dV$$

and

$$P_{\text{hyst}} = \int_V \frac{1}{2} \omega \mu'' H^2 dV,$$

where σ is the electrical conductivity, E is the electric field intensity, H is the magnetic field strength, and μ'' is the imaginary part of the complex permeability [13], [28]. Because the electrical conductivity of Mn-Zn ferrite is much lower than that of metallic shielding materials, the eddy-current contribution is strongly suppressed at low frequency [19], [20]. In the frequency range considered in this work, the hysteresis loss is therefore used as the dominant noise source. The magnetic-noise amplitude can be approximated as

$$\delta B_{\text{hyst}}(f) = \frac{1}{AI} \left(\frac{2kT\mu''}{\pi f} \int_V H^2 dV \right)^{1/2}.$$

Thus, when the material parameters and frequency are fixed, the quantity

$$\text{IntH2} = \int_V H^2 dV$$

is used as the finite-element indicator of magnetic-noise contribution. For a multilayer ferrite shield, the total indicator is calculated as

$$\text{IntH2}_{\text{total}} = \sum_{i=1}^{N_L} \text{IntH2}_{L_i},$$

where N_L is the number of ferrite layers and IntH2_{L_i} is the layer-resolved integral of the i th layer. This layer-resolved form is used to identify whether the innermost layer or the outer layers dominate the total magnetic-noise contribution.

B. Multilayer Cylindrical Ferrite-Shell Geometry

The investigated shield is a cylindrical ferrite structure composed of N_L concentric thin layers. Each layer has a ferrite thickness a , and adjacent layers are separated by a radial gap g , which represents air or adhesive spacing. Similar multilayer and gapped magnetic-shield configurations have been shown to require explicit geometric modeling because small discontinuities can change both shielding performance and magnetic-noise distribution [29]-[31]. The total radial occupation thickness is

$$T_{\text{space}} = N_L a + (N_L - 1)g.$$

When the total radial space and gap are fixed, the ferrite thickness of each layer is

$$a = \frac{T_{\text{space}} - (N_L - 1)g}{N_L}.$$

This relationship enables a fair comparison among different layer numbers under the same radial design envelope. In the manufacturable thin-sheet study, the single-layer ferrite thickness is constrained to

$$0.08 \text{ mm} \leq a \leq 0.60 \text{ mm}.$$

The continuous-shell model is used as the first-level screening model. It represents each ferrite layer as an ideal closed cylindrical shell and is used to evaluate the influence of layer number, ferrite thickness, radial gap, and material volume. After the promising continuous-shell candidates are identified, a segmented-shell model is used as the second-level correction model. In the segmented model, the cylindrical surface is divided into ferrite strips along the circumferential direction, and the circumferential gap between adjacent strips is explicitly included. The segmentation correction is described by the number of strips N_{seg} , the circumferential gap g_ϕ , and the coverage fraction η_ϕ .

C. Finite-Element Calculation of Shielding Factor

The static shielding performance is evaluated using ANSYS Maxwell finite-element simulations. Finite-element analysis has been widely used for magnetic shields with apertures, gaps, winding structures, and nonlinear magnetic materials because closed-form analytical solutions are difficult to obtain for practical three-dimensional structures [29]-[35]. For each structure, two models are calculated under the same external magnetic-field excitation. In the reference model, the ferrite material is replaced by vacuum, and the center magnetic flux density is recorded as B_0 . In the shielded model, the ferrite

layers are retained, and the center magnetic flux density is recorded as B_{center} . The shielding factor is defined as

$$\text{SF} = \frac{|B_0|}{|B_{\text{center}}|}.$$

The residual ratio is defined as

$$\text{ResidualRatio} = \frac{|B_{\text{center}}|}{|B_0|}.$$

For three-axis evaluation, the center-field components B_x , B_y , and B_z are exported in both the reference and shielded models. The total center-field magnitude is calculated as

$$B = \sqrt{B_x^2 + B_y^2 + B_z^2}.$$

The same reference field and observation point are used for all cases to ensure that the shielding factors are directly comparable. The material efficiency of each candidate is evaluated by the logarithmic shielding efficiency per ferrite volume,

$$\eta_S = \frac{\ln(\text{SF})}{V_f},$$

where V_f is the total ferrite volume. This metric is used to compare structures with different total material usage.

D. Finite-Element Calculation of Magnetic-Noise Indicator

The magnetic-noise indicator is calculated by applying a virtual excitation coil at the shield center and integrating the squared magnetic field strength over the ferrite material. The excitation current is kept identical for all cases, so IntH2 can be compared directly among different geometries. Low-frequency magnetic-noise measurements can also be affected by environmental vibration and sensor $1/f$ noise; therefore, this simulation indicator is used as a geometry-dependent source metric rather than as a replacement for final experimental noise characterization [36], [37]. For a single-layer shell, the integral is

$$\text{IntH2}_{\text{total}} = \int_{V_f} H^2 dV.$$

For a multilayer shell, each ferrite layer is retained as an independent solid body in the finite-element model. The layer-resolved integrals are exported as

$$\text{IntH2}_{L_i} = \int_{V_{L_i}} H^2 dV, \quad i = 1, 2, \dots, N_L.$$

The total magnetic-noise indicator is then obtained by summation. This treatment avoids merging all ferrite layers into a single body and allows the contribution of each layer to be analyzed separately. The innermost-layer value IntH2_{L_1} is reported together with $\text{IntH2}_{\text{total}}$, because the innermost layer is closest to the protected volume and can have a direct influence on the center magnetic noise.

E. Continuous-Shell Screening and Segmented-Shell Correction

The simulation workflow contains two levels. First, continuous cylindrical shells are used to screen the main structural parameters. The screened variables include layer number N_L , ferrite thickness a , radial gap g , total radial thickness T_{space} , and total ferrite volume V_f . The exported quantities for each case are B_0 , B_{center} , SF, ResidualRatio, IntH2_{L_i} , and $\text{IntH2}_{\text{total}}$. This stage is used to determine whether multilayering remains beneficial when each ferrite layer is limited to the manufacturable thickness range.

Second, selected candidates are rebuilt as segmented three-dimensional shells. In this stage, circumferential slots between ferrite strips are included explicitly. The segmented model quantifies the penalty introduced by noncontinuous ferrite coverage, which is unavoidable in a sheet-based prototype. The continuous-shell and segmented-shell results are compared using the same shielding and noise metrics. A multilayer design is considered favorable only when it satisfies the following conditions:

$$\text{SF}_{\text{multi}} \geq \text{SF}_{\text{target}},$$

$$V_{\text{multi}} < V_{\text{single}} \quad \text{at comparable SF},$$

and

$$\text{IntH2}_{\text{total,multi}} \leq \text{IntH2}_{\text{limit}}.$$

This decision rule ensures that the multilayer structure is not judged only by shielding factor, but also by ferrite consumption and magnetic-noise contribution.